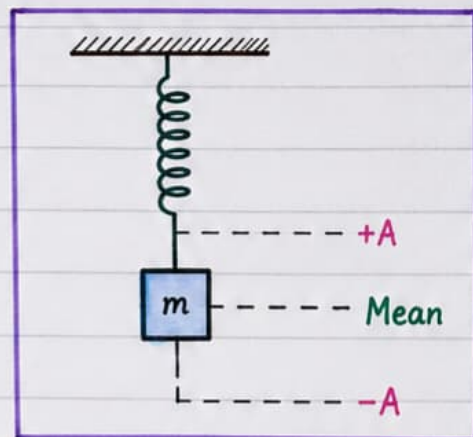


Oscillations

What is Oscillation?

The to and fro or back and forth motion of a body about a fixed mean position is called **oscillatory motion** or **oscillation**.



Types of Oscillations :

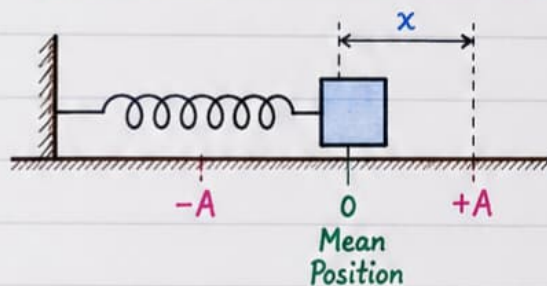
1. **Free Oscillations** : When a body oscillates freely under the influence of restoring force alone after being disturbed.
2. **Forced Oscillations** : When a body oscillates under the influence of an external periodic force.

Key Points

- Oscillation is periodic motion.
- It repeats itself after equal intervals of time.
- It occurs due to restoring force.

Displacement in SHM :

Displacement (x) is the shortest distance of the oscillating body from the mean position at any instant.



Important Terms :

- **Mean Position** : The central position about which the body oscillates.
- **Amplitude (A)** : Maximum displacement from the mean position.
- **Time Period (T)** : Time taken to complete one oscillation.
- **Frequency (ν)** : Number of oscillations per second. $\nu = 1/T$
- **Angular Frequency (ω)** : $\omega = 2\pi\nu = \frac{2\pi}{T}$

Simple Harmonic Motion (SHM) :

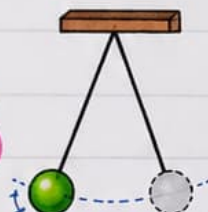
If the **restoring force** is directly proportional to the displacement and acts towards the mean position, the body performs SHM.

$$F \propto -x \Rightarrow F = -kx$$

where, k = force constant

Examples :

- Mass-spring system
- Simple pendulum (small angle)
- Torsional pendulum



★ In ideal SHM, there is no loss of energy due to friction or air resistance.



Chapter - 13 : Oscillations

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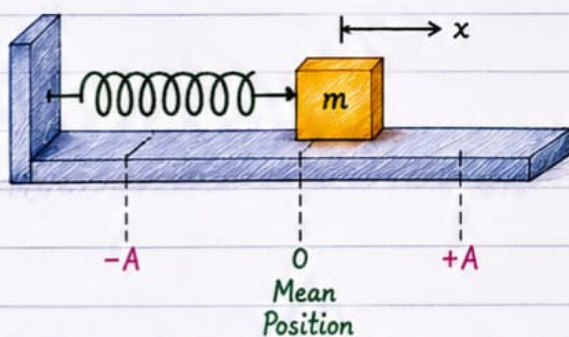
1. Simple Harmonic Motion (SHM):

A motion in which the restoring force (F) is directly proportional to the displacement (x) from the mean position and acts towards it is called **SHM**.

$$F \propto -x \Rightarrow F = -kx$$

where, k = force constant ($N m^{-1}$)

Spring - Mass System (Example of SHM)



2. Restoring Force and Force Constant :

- Restoring force tries to bring the body back to the mean position.
- Larger $k \Rightarrow$ stronger restoring force.

3. Time Period (T):

- Time taken by the body to complete one oscillation is called time period.
- SI unit : **second (s)**

Time Period in SHM

(i) Spring-Mass System

$$T = 2\pi \sqrt{\frac{m}{k}}$$

where,

m = mass (kg)

k = force constant ($N m^{-1}$)

(ii) Simple Pendulum

$$T = 2\pi \sqrt{\frac{l}{g}}$$

where,

l = length (m)

g = acceleration due to gravity ($\approx 9.8 m s^{-2}$)

4. Amplitude (A):

Maximum displacement of the body from the mean position.

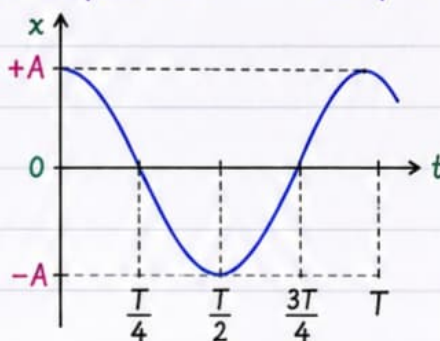
5. Frequency (ν):

Number of oscillations completed per second.

$$\nu = \frac{1}{T}$$

More ν means more oscillations, per second.

6. Displacement - Time Graph in SHM



- At $t = 0$, $x = +A$
- At $t = \frac{T}{4}$, $x = 0$
- At $t = \frac{T}{2}$, $x = -A$
- At $t = \frac{3T}{4}$, $x = 0$
- At $t = T$, $x = +A$

7. Comparison : SHM vs Non-SHM

Feature	SHM	Non-SHM
Restoring Force	$F \propto -x$ ($F = -kx$)	Not proportional to $-x$
Path	Periodic and repeats after equal intervals	Not necessarily periodic
Time Period	Constant (for a given system)	May vary
Energy	Total mechanical energy is constant (ideal case)	May change due to non-conservative forces
Example	Spring-mass, Simple pendulum (small angle)	Motion of a car in traffic, motion on a rough surface

Energy in SHM

Total mechanical energy (E) of a body executing SHM is constant.

$$E = \frac{1}{2} k A^2$$

where,

A = amplitude

k = force constant



For small oscillations, many complex motions can be approximated as **SHM**.



Chapter - 13 : Oscillations

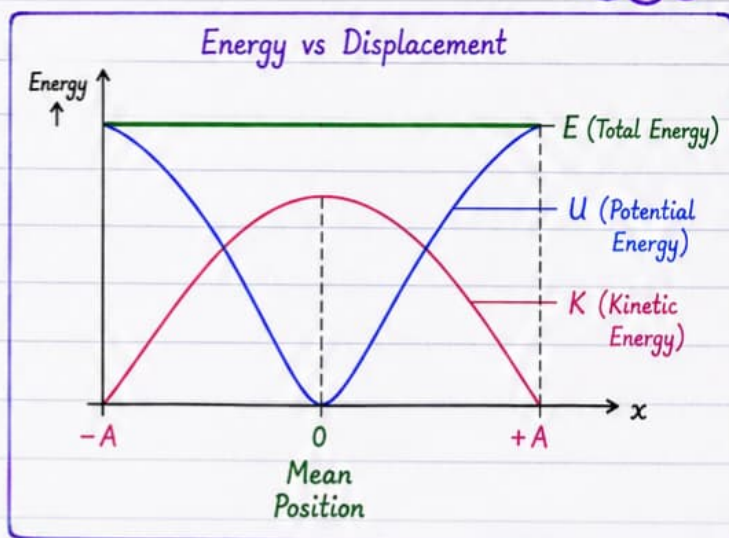
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8. Energy in SHM :

- In SHM, total mechanical energy (E) of the system remains constant (in absence of friction).
- It is the sum of kinetic energy (K) and potential energy (U).

$$E = K + U = \text{constant}$$

- ★ At mean position ($x = 0$)
 K is maximum, U is minimum (zero).
 At extreme positions ($x = \pm A$)
 U is maximum, K is minimum (zero).



9. Phase and Phase Constant :

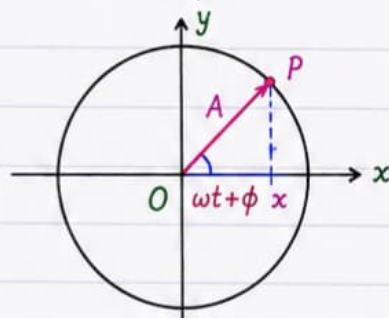
- The state of the oscillator at any instant is described by its phase (ϕ).
- It is the argument of the sine or cosine function used in the equation of SHM.
- General equation of SHM :

$$x = A \sin(\omega t + \phi)$$

where, ϕ = phase constant (depends on initial conditions)

- At $t = 0$, the phase of the oscillator is ϕ .

Phasor Representation



Projection of OP on x -axis gives displacement x .

10. Differential Equation of SHM :

- For a particle executing SHM, acceleration is directly proportional to displacement and acts towards the mean position.

$$a \propto -x \Rightarrow a = -\omega^2 x$$

or,

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \Rightarrow \text{Differential equation of SHM}$$

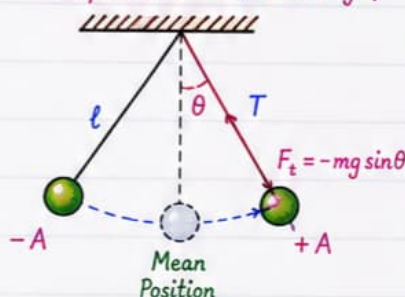
- x = displacement
- a = acceleration
- ω = angular frequency (rad s^{-1})
- Negative sign shows that acceleration is opposite to displacement (restoring nature).



11. Simple Pendulum :

- A simple pendulum consists of a small heavy bob suspended by a light, inextensible string.
- When displaced through a small angle (θ) and released, it executes SHM.

Simple Pendulum (Small Angle)



For small angles,
 $\sin \theta \approx \theta$ (in radians)

Time period :

$$T = 2\pi \sqrt{\frac{l}{g}}$$

where,
 l = length of pendulum
 g = acceleration due to gravity



SHM is a powerful model to describe many natural and practical systems.



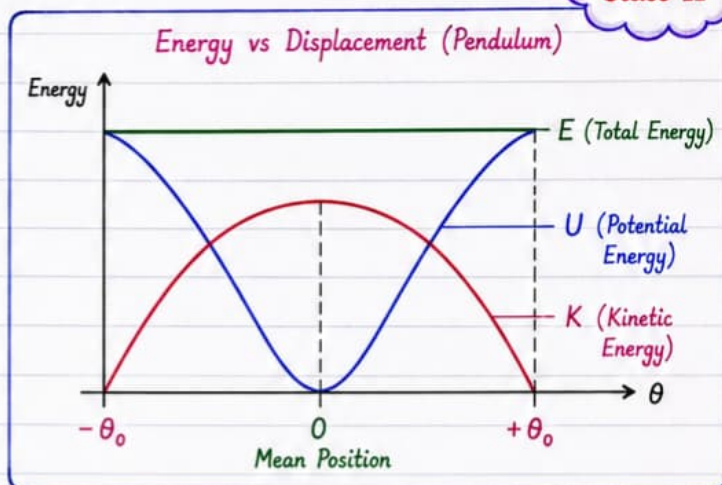
Chapter-13 : Oscillations

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12. Energy in Simple Pendulum :

- The total mechanical energy (E) of the pendulum remains constant.
- At mean position ($\theta = 0$) : K is **maximum**, U is **minimum** (zero).
- At extreme positions ($\theta = \pm\theta_0$) : U is **maximum**, K is **minimum** (zero).

★ $E = K + U = \text{constant}$



13. Compound (Physical) Pendulum :

- A rigid body free to oscillate about a fixed horizontal axis **not passing through** its **centre of mass**.
- Time period for small oscillations :

$$T = 2\pi \sqrt{\frac{I}{Mgh}}$$

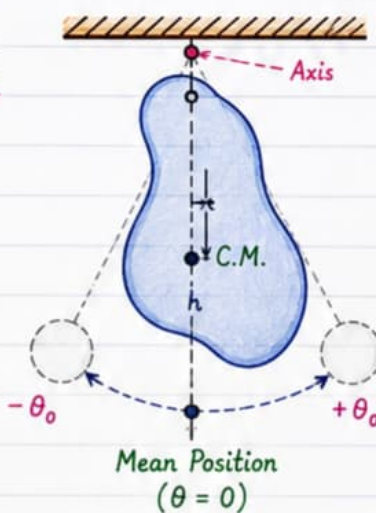
where,

I = moment of inertia about the axis

M = mass of the body

h = distance between axis and centre of mass

g = acceleration due to gravity



Special cases :

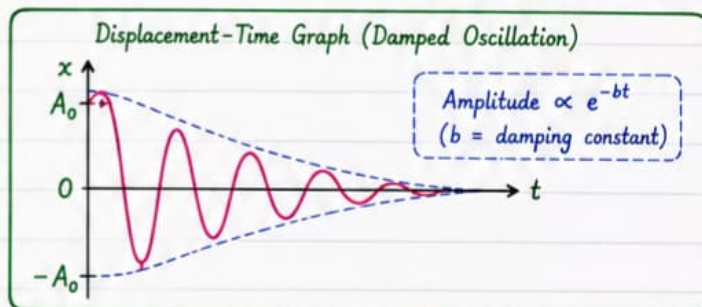
- If axis passes through C.M. $\Rightarrow h = 0 \Rightarrow T = 0$ (no restoring torque).
- If axis is far from C.M. (h large) $\Rightarrow T$ increases.

It behaves like a simple pendulum of length

$$L_{\text{eff}} = \frac{I}{Mh}$$

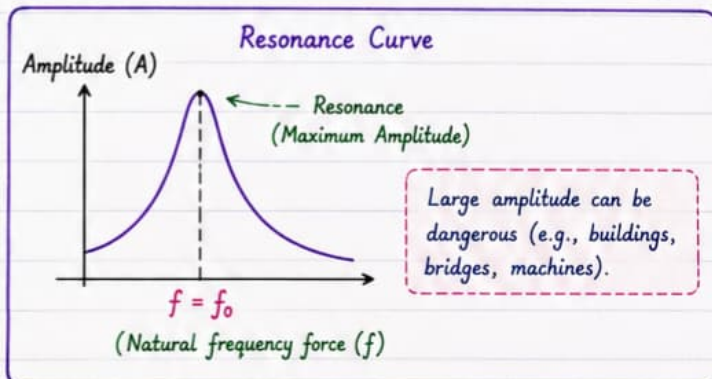
14. Damped Oscillations :

- When resistive forces like friction or air resistance act on the oscillating body, the **amplitude decreases with time**.
- The body still executes oscillations but the energy is gradually lost to the surroundings.



15. Forced Oscillations :

- When a periodic external force acts on the system, it oscillates with the same frequency as the external force.
- If the frequency of the external force equals the natural frequency of the system, **resonance** occurs and the amplitude becomes maximum.



In Summary

- ✓ Oscillation is periodic to-and-fro motion.
- ✓ SHM is the ideal case with linear restoring force.
- ✓ Energy in SHM remains constant.
- ✓ Many systems (pendulum, spring, etc.) show SHM.
- ✓ Damping decreases amplitude; resonance increases it.

♥ Understanding **oscillations** helps us explain and design many natural and engineering systems around us.